

Rahul Joban

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Full-Stack Software Engineer with 6+ years building customer-facing web products. Proven track record delivering React/Node.js applications on AWS that drive measurable user growth. Experienced across the full lifecycle; from database design and REST APIs to polished UIs and production deployments.

Technical Skills

Frontend: React, Next.js, JavaScript (ES6), HTML5, CSS3

Backend: Node.js, FastAPI, Java, Python, SQL, REST APIs

Cloud & Tools: AWS (EC2/S3), Azure, Git, Agile

Other: C/C++(Linux/Unix), ML/CV (PyTorch, Azure Computer Vision), AI Native Workflow (GitHub, Copilot, ChatGPT), embedded systems, Top Secret Clearance, Spanish Fluency, Technical Documentation

Projects

Robokidds Coding Academy- Full-Stack Student Platform, UX design | React, Node.js, AWS, Azure, SQL

- Built student portal with React frontend and cloud infrastructure on AWS, **increasing school enrollment by 40%**.
- Implemented facial recognition login using Microsoft Azure AI Foundry Tools.
- Designed **SQL database** to track billing and attendance for 200+ students.
- Taught web design, robotics, and programming to K-12 students.

SSPA Firmware & Test Automation | C/C++, Python, Embedded Systems, Data Visualization

- Developed production firmware for space-grade amplifiers with rigorous hardware validation.
- Created automated test framework with real-time telemetry visualization for customer presentations.
- Integrated with signal generators to test communications across frequency ranges.

Professional Experience

Full-Stack Software Engineer

July 2022 – Present

Arka Group, LP

- Lead Software Engineer for Solid State Power Amplifiers (SSPA) used in **Blue Origin lunar lander** and **NASA constellation satellites** (75+ units delivered).
- Built high-throughput data ingestion pipelines for telemetry data, pattern directly applicable to content ingestion systems at scale.
- Built Python-based automated test suite with a GUI used by **15+ engineers**, reducing debug time by **60%**.
- Developed internal **data visualization** dashboards for real-time telemetry from aerospace hardware.
- Authored **technical documentation** and led **Agile reviews** for cross-functional teams.
- Collaborate with electrical engineers to interpret schematics and validate hardware using **oscilloscopes, multimeters, and logic analyzers**.
- Architect and implement embedded C/C++ firmware on TI MSP430 microcontrollers, controlling Digital Step Attenuators (DSAs), DACs/ADCs via SPI/I2C, GPIO, Timers, and UART.

Embedded Software Engineer

July 2019 – July 2022

Pacific Defense – Seattle, WA

- Developed embedded Linux features and GUI tools for Software Defined Radios (SDRs) in C/C++ and Python for military applications.
- Built and maintained **client APIs** for MicroScan SDR, supporting Department of Defense customers.
- Created comprehensive unit tests that **increased test coverage and reduced user-reported issues by 45%** within one month.
- Obtained and maintained **Top Secret Security Clearance**.

Education

University of British Columbia

Relevant Coursework: Data Structures, Algorithms, Software Engineering, Web Systems, Computer Security, UX Development